Course Content and Resources

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1 The Eight Learning Objective Bundles

1.1 The Four-Step Problem Solving Method

There is only one learning objective for this bundle, and it's properly using the four-step method when solving a problem. If you do all four steps below, you will receive credit for this learning objective on relevant problems.

1. Draw a diagram
2. Gather your information – choose a coordinate system, identify "knowns" in the problem statement (which are allowed to be in your answer), list "unknowns" which must be solved for or eliminated, and determine your goal (what quantity you are solving for and any special conditions for it)
3. Solve symbolically: list key concepts that may guide your problem solving, find appropriate equations for solving problems using those key concepts, and use those equations and math tools to eliminate all unknowns and lead to your answer. Do not plug in numbers.
4. Assess your answer. Check units and use some logical reasoning to defend your answer (or explain why it's wrong). Some methods you could use for the reasoning includes making predictions and seeing if your result matches those predictions, relating the answer to something you know (e.g. plugging in specific numbers and checking if they match real-world scenarios), and evaluating the equation (check limits and special cases, make sure it does what you expect)

1.2 Using Math Tools

A lot of problem solving in this class requires applying mathematical tools you've learned in other classes. Here are the various learning objectives you may see that involve using math tools

- Extremize a function by taking a derivative
- Take a limit to determine how an expression changes as parameters get large
- Use a Taylor series to determine how an expression changes as parameters get small
- Determine an angle of a right triangle
- Determine the length of a side of a right triangle
- Use a trig identity to simplify an expression
- Set up an integral by determining the differential components
- Calculate an indefinite integral
- Calculate a definite integral
- Solve an ODE by separating the variables and integrating
- Solve a system of N equations (or inequalities) with N unknowns
- Properly set up the coordinate system for a system using polar coordinates
- Convert between coordinate systems

1.3 Forces and Torques

Here are the learning objectives you may see that involve forces and torques:

- Identify the forces acting on an object

- Construct a free body diagram that accounts for all forces on an object
- Resolve a force into components in your chosen coordinate system
- Use inequalities to analyze static friction conditions
- Identify whether static friction is maximized
- Balance forces on a static object
- Calculate the torque from a force relative to a pivot
- Construct a torque diagram with a chosen pivot
- Balance torques on a static object
- Apply Newton's second law to connect the forces on an object to its acceleration
- Construct an interaction diagram that accounts for forces between objects in your system and external forces
- Apply Newton's third law to find forces between objects in a multi-object system

1.4 Motion

Here are the learning objectives you may see that involve motion and kinematics:

- Construct the equations of motion using N2L
- Calculate the velocity of an object from its acceleration vector
- Calculate the position of an object from its velocity vector
- Derive or interpret the expressions for acceleration in polar coordinates
- Derive or interpret the expressions for velocity in polar coordinates
- Calculate kinematic variables (distance traveled, time, final velocity, launch angle, etc) given equations of motion
- Use or explain the decoupling of x and y motion in projectile motion

1.5 Work and Energy

Here are the learning objectives you may see that involve work and energy:

- Define a central force
- Explain the role of conservative forces in energy conservation
- Apply the law of universal gravitation to calculate the potential energy of spherical masses
- Apply Hooke's Law to calculate the potential energy of masses on springs
- Calculate the kinetic energy of an object
- Calculate the potential energy of an object due to a spring force
- Distinguish between kinetic, potential, and total mechanical energy
- Determine whether mechanical energy is conserved
- Recognize turning points and forbidden regions using potential energy diagrams
- Calculate a position or speed using the conservation of mechanical energy equation
- Calculate the work done by a force
- Calculate a force, displacement, or speed using the work-energy theorem

1.6 Momentum and Impulse

- Determine whether momentum is conserved
- Apply the impulse momentum theorem to a system where momentum is not conserved
- Explain the connection between Newton's Third Law and momentum conservation
- Apply momentum conservation and/or $F = dp/dt$ in systems with changing mass
- Define and interpret the center of mass (CM) frame
- Calculate the velocity of the CM frame for a multi-particle system
- Explain why momentum conservation holds in all inertial frames
- Transform between the lab frame and the center of mass frame
- Determine whether a collision is elastic or inelastic
- Apply conservation of momentum and kinetic energy to solve elastic collision problems
- Apply conservation of momentum to solve inelastic collision problems
- Construct the Lagrangian for a system
- Determine the equations of motion from the Lagrangian
- Calculate the kinetic energy of a rigid body undergoing translation and rotation

1.7 Angular Momentum

- Calculate the angular momentum of a point particle
- Calculate the angular momentum of a rigid body undergoing translation and rotation
- Determine whether angular momentum is conserved
- Use the effective potential to determine turning points, bound vs. unbound motion, and qualitative orbital behavior
- Relate energy, angular momentum, and eccentricity, and use these to classify orbits (circular, elliptical, parabolic, hyperbolic)
- Use angular momentum conservation to relate different points along a trajectory
- Calculate moments of inertia by summing point-mass contributions or integrating over a continuous mass distribution
- Use the perpendicular-axis theorem to calculate a moment of inertia
- Use the parallel-axis theorem to calculate a moment of inertia
- Explain and apply the center-of-mass decomposition of angular momentum and kinetic energy
- Define the effective potential for a system
- Use the effective potential to determine regions of bound and unbound motion

2 Math and Physics Resources

2.1 External Resources

Khan Academy has wonderful videos not just on the math you need for this class, but also physics from previous courses. You can find their physics archive here.

2.2 The Four-Step Problem Solving Method

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2. Gather your information – choose a coordinate system, identify "knowns" in the problem statement (which are allowed to be in your answer), list "unknowns" which must be solved for or eliminated, and determine your goal (what quantity you are solving for and any special conditions for it)
3. Solve symbolically: list key concepts that may guide your problem solving, find appropriate equations for solving problems using those key concepts, and use those equations and math tools to eliminate all unknowns and lead to your answer. Do not plug in numbers.
4. Assess your answer. Check units and use some logical reasoning to defend your answer (or explain why it's wrong). Some methods you could use for the reasoning includes making predictions and seeing if your result matches those predictions, relating the answer to something you know (e.g. plugging in specific numbers and checking if they match real-world scenarios), and evaluating the equation (check limits and special cases, make sure it does what you expect)

2.3 The Euler-Lagrange Equations

The Euler–Lagrange equations help us find how a system moves. They come from the idea that nature chooses the path that makes the "action" (a quantity built from kinetic and potential energy) as small as possible. Instead of focusing directly on forces, these equations let us work with energy and generalized coordinates, which often makes solving problems—especially with constraints—much simpler.

All the equations we will use this semester arise from the Euler-Lagrange equations, from energy conservation to Newton's second law to conservation of momentum.

To use them, you have to construct the Lagrangian (L = T -V), determine the proper coordinates to use (i.e. what coordinates will you use to describe your problem and your solution?), and then plug them into the Euler-Lagrange equation, which involves taking derivatives of the Lagrangian with respect to the coordinates you've chosen. After simplifying the results, you will have the equations of motion for the system.

2.4 Math Review

2.4.1 Equations & Inequalities

Solving systems of equations

Systems of inequalities

2.4.2 Exp & Log

Logarithm properties

Exponent properties

2.4.3 Vectors & Trig

Vector review

The cross product

Trig equations and identities

Trig cheat sheet

2.4.4 Extremes

Intro to Taylor series

List of common series

Finding local mins and maxes

Taking limits

2.4.5 Coordinate Systems

Polar Coordinates

2.4.6 Integrals & ODEs

The separate and integrate method

Common integrals and derivatives